



“Not all convenience stores are equal”. An analysis of the criminogenic role of corner Stores in Newark, NJ

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ABSTRACT

This study investigates the spatial relationship between corner stores and crime in Newark, New Jersey, to assess whether these venues play a distinct criminogenic role compared to other businesses. Although previous literature has often treated convenience stores as a homogeneous risk category, we hypothesize that corner stores exert a unique influence on crime patterns. Using crime data for 2022, this paper aims to investigate how the distribution of gun violence, robberies, and aggravated assaults is spatially connected to proximity to corner stores in comparison with other similar venues, like convenience stores, pharmacies, retail stores, restaurants, or gas stations, and how these associations are influenced by neighborhood characteristics and temporal variables. Our findings reveal that crime is significantly concentrated around corner stores, and that this association is stronger for gun violence and aggravated assaults than for robberies. Interaction effects suggest that neighborhood characteristics shape the degree and nature of crime around these stores. Further, crime persists around corner stores during both day and night, implying mechanisms beyond mere business hours. The study underscores the need to disaggregate place-based risk assessments and to differentiate interventions considering both micro-place features and broader contexts. These insights have implications for crime prevention, urban planning, and community safety policies.

1. Introduction

The analysis of the connection between crime and the surrounding environment is crucial for informing crime prevention strategies. A substantial body of literature argues that addressing the characteristics of places that facilitate criminal behaviors is more likely to yield significant and lasting impacts on crime levels than merely focusing on the repression of crime events themselves (Kennedy et al., 2018; Weisburd, 2008). However, this process is often impeded by the challenge of correctly identifying the criminogenic links that connect context and crime. This challenge arises from the multitude of potential elements or factors that characterize specific places and the inherent complexity of the interactions among them (Wikström, 2011). One way to partially mitigate this problem is to adopt analytical strategies that aim to model and incorporate this complexity as much as possible. This implies that any analysis should always be paired with clear assumptions of the specific criminogenic influence of the elements considered in the social

and physical environment. The aims are, on the one hand, to effectively guide more tailored analyses and, on the other hand, to obtain results that could be more easily translated into a practical understanding of police officers or other stakeholders (Kennedy et al., 2021). For example, it should be acknowledged that even similar environmental factors may be differently linked to criminal behaviors or security issues according to specific and diverse peculiarities.

Starting from these premises, the extant paper focuses on a practical case study to highlight how different types of convenience stores are dissimilarly linked to crime. The base of this research was the results of a previous analysis conducted by the Newark Public Safety Collaborative (NPSC), a Rutgers University-Newark Anchor initiative launched in 2018 in the city of Newark, NJ, and housed at the Rutgers School of Criminal Justice. This previous analysis highlights that gun violence incidents in Newark largely concentrate around a subset of small and often family-owned convenience stores, commonly known as corner stores or *bodegas* (Barrett & McWhirter, 2023). These results were not

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surprising, considering that convenience stores have often been highlighted by previous studies as potential crime generators or attractors (Askey et al., 2018; D'Alessio & Stolzenberg, 1990; Duffala, 1976). However, in the discussion prompted by these findings among the NPSC community partners, many of the emerging suggestions from local stakeholders focused on the specificity of corner stores and *bodegas* and their social role for local communities to explain their potential criminogenic mechanisms and the observed association. These insights support the assumption that corner stores should be considered separately from other convenience stores or similar businesses, as they may play different social and economic roles in local communities, and thus, they may affect criminal behaviors differently.

Expanding on these findings, this paper aims to explore whether the criminogenic role of corner stores is significantly different from those associated with other types of convenience stores. Indeed, convenience stores and similar establishments are considered a single type of crime generators or attractors in many environmental criminology studies (e.g., see Connealy, 2021; Haberman & Ratcliffe, 2015; Tillyer et al., 2021). The assumption is that, despite their uniqueness, all those places have more communalities than differences and, therefore, could be considered as a homogeneous category of risky facilities for explaining criminal behaviors (Askey et al., 2018; Eck et al., 2009). However, some studies have already concluded that, given their specificities, different types of convenience stores may have different connections with criminal behaviors. For example, a previous study demonstrates that independent convenience stores are more likely to suffer a robbery than company-owned ones (Hendricks et al., 1999), while another research establishes that violent crimes concentrate slightly more around franchise stores than corner stores (Furr-Holden et al., 2016). Thus, this paper aims to investigate how the likelihood of different criminal events is spatially connected to the presence of corner stores in comparison with other similar venues, like convenience stores, pharmacies, or gas stations, and how these associations are influenced by other background or temporal variables.

The paper is organized as follows. The first part summarizes existing knowledge on the role of convenience stores as risk factors for crimes and introduces the peculiarities of corner stores in Newark. The second section describes the hypotheses, the methodology, and the data used in the analysis. The third part presents the results, while the last section discusses the main findings and draws conclusions.

2. Convenience stores and crime

The relationship between convenience stores and crime is well established, supporting that these venues can both attract and generate a wide range of criminal activity (Kennedy et al., 2022). For example, research has indicated that convenience stores correlate with violent crime incidents (Connealy, 2020; Furr-Holden et al., 2016; Singleton et al., 2019; Valasik et al., 2018; White & Muldoon, 2015). Whereas, other studies indicate that they are frequently located in areas identified as robbery hotspots (Connealy & Piza, 2019; Hatten & Piza, 2022). In general, the existing literature is fairly consistent in highlighting the role of convenience stores in contributing to higher crime rates or the concentration of antisocial conduct in their vicinity.

This association can be explained by the situational crime opportunities that arise from the inherent nature of convenience stores, primarily linked to cash-heavy transactions, extended operating hours, and the social dynamics fostered by these venues within the surrounding neighborhoods (Altizio & York, 2007; Duffala, 1976). On the one hand, convenience stores are often targeted for property crimes against them or their clients, due to the availability of valuable or accessible targets, such as cash or other small items (Gotham & Kennedy, 2021). On the other hand, some scholars highlight the lack of effective guardianship due to their late-night hours, rapid customer turnover, and limited supervision that can generate illegal activity by attracting disorderly behavior (Anderson, 2000; Calder & Bauer, 1992; Wellford et al., 1997;

White & Muldoon, 2015).

Convenience stores can also attract crime due to their role within an environmental context. Many of these businesses are located in low-income neighborhoods, where socio-economic challenges such as poverty and limited access to community resources can exacerbate crime rates (Hendricks et al., 1999). The presence of convenience stores in these areas can create a dual effect: while they provide essential goods and services (Ruff et al., 2016), they can also attract and concentrate criminal activities (D'Alessio & Stolzenberg, 1990). According to these views, the impact of convenience stores on crime may be indirect, as they attract many people who rely on them to purchase essential food items and other household supplies, particularly in socially disorganized and disadvantaged urban settings where large supermarkets are less present or accessible. As a result, they often become local service hubs and gathering points for community interactions (Werner, 2023).

Both these effects can be further intensified by the specific location of convenience stores, particularly those near major roads, gas stations, public transport, or in areas with higher human activity, that may increase crime risk (Barnum, Caplan, et al., 2017; Calder & Bauer, 1992; Tillyer & Walter, 2019). The high volume of foot traffic of people who regularly visit or pass through these establishments increases the likelihood of encounters between potential victims and perpetrators, resulting in a higher concentration of deviant behaviors or crimes around convenience stores (Scott, 2011; White & Muldoon, 2015). For example, Shin (2024) highlights that affordable convenience stores in socially disadvantaged areas, often open 24 h, attract a demographic that is statistically more likely to be involved in violent crime, particularly young males. Furthermore, these same factors make convenience stores suitable places to serve as sites for more organized criminal activities, including drug sales or the fencing of stolen goods (Barnum, Campbell, et al., 2017; Clarke & Eck, 2007; Gruenewald, 2013). Another factor influencing crime in and around convenience stores is the sale of alcohol, which can exacerbate the risk of violence and disorder. Research has consistently found that areas with a higher density of alcohol-selling establishments experience more violent crimes (Gruenewald et al., 2006; Snowden, 2021; Toomey et al., 2012). Convenience stores that sell alcohol can contribute to this trend by increasing the likelihood that intoxicated people engage in violent or disorderly conduct.

Finally, the specific characteristics of the store design and management can influence criminal behaviors within and around the venue. Place managers, such as store owners or employees, community organizers, or residents can play a vital role in fostering a safer environment by implementing crime prevention strategies and engaging with local law enforcement (Altizio & York, 2007; Clarke & Eck, 2007; Linning & Eck, 2021). The absence of effective or vigilant place managers, weak social controls, or lack of community cohesiveness, along with differences in internal policies, store design, or security measures, can help explain variations in crime rates across convenience stores (Exum et al., 2010; Kajalo & Lindblom, 2011; Mazerolle et al., 1998). The role of place managers or business owners is particularly relevant, as they possess the authority and capacity to create or regulate criminal opportunities within the establishments they oversee and their immediate surroundings (Linning et al., 2022). Although their effectiveness in preventing crime largely depends on the broader social and physical context in which these places are embedded (Tucker & O'Brien, 2024).

3. Corner stores and crime in Newark

In Newark, New Jersey corner stores or *bodegas* are an integral part of the urban landscape, serving as community anchors that provide convenient access to essential goods at affordable prices. Often located on street corners, hence the name, these small, independently owned neighborhood convenience stores offer everyday essentials such as snacks, beverages, canned goods, loosies (single cigarettes), household supplies, deli items, and occasionally, fresh produce. In Spanish-

speaking communities throughout Newark, New York City, and other urban areas, these small shops, commonly known as *bodegas*, offer culturally specific products tailored to Latino and Hispanic residents. The term *bodega* translates from the Spanish word for “warehouse” or “small wine cellar,” but in this context, it refers to small neighborhood shops. Both corner stores and *bodegas* are deeply embedded in urban areas with roots in the community, operate on a small footprint, and are easily accessible on foot.

Unlike branded convenience stores, such as those located in gas stations or national convenience store chains, corner stores are typically independently owned and deeply connected within the neighborhoods they serve. Store owners and staff often build long-standing relationships with local customers, positioning them as informal community hubs (Dombrowski & Kelley, 2019; Mayer et al., 2016). In contrast, chain-affiliated convenience stores tend to have higher staff turnover and offer goods at higher prices than locally owned corner stores. This difference is relevant also considering previous studies stressing the role of local ownership in determining the criminogenic role of specific businesses. In particular, scholars claimed that business activities that are not owned by non-local residents are more likely to concentrate criminal activities, because the owners or employees have less interest or have lower social resources to intervene in preventing crime (Kim & Hipp, 2022; Linning & Eck, 2021).

In addition to their convenience, corner stores play a critical role in the daily lives of residents, especially in historically underserved and low-income neighborhoods where they tend to concentrate. In Newark, branded convenience stores tend to be widely spread across the city and do not present the spatial clustering within disadvantaged neighborhoods, while corner stores are commonly found in socially disorganized areas, particularly in parts of the South and West Wards. These neighborhoods have historically experienced high levels of crime and show visible signs of urban decay. The role played by corner stores in these areas is ambivalent.

On the one hand, corner stores extend beyond their commercial function and represent for many residents more than just places of economic transactions. Community participants at an NPSC Community Stakeholders’ meeting shared how these places play a key role in supporting local communities and tend to become a regular meeting point for residents to interact and socialize. Local community leaders also discussed how residents rely on neighborhood corner stores to get food and basic supplies. This is not surprising, as in many communities, especially those within the so-called “food deserts”, corner stores may be the only source of food within several miles (Kennedy et al., 2024). Further, corner stores and *bodegas* often carry culturally specific goods that reflect the ethnic composition of the surrounding area.

On the other hand, given their location within these neighborhoods, extended hours, and relatively low levels of formal surveillance compared to other businesses, these establishments can become attractive spaces for low-level drug transactions and other illicit activities. They often serve as informal public spaces where young people and other individuals gather which can create opportunities for crime and disorder to emerge and persist (Scott, 2011). In some cases, these gatherings have also been associated with more serious acts of violence, including homicides and drive-by shootings (Cohen, 2022).

Importantly, these establishments are not necessarily the root of the problem. Rather, their criminogenic characteristics can be shaped by the broader conditions surrounding these places, such as concentrated poverty, limited economic opportunities, and long-term disinvestment in urban infrastructure. In many cases, owners and employees of these stores find themselves in difficult situations, unable to prevent these activities from occurring near their businesses, despite having a vested interest in improving the conditions around them. In other cases, however, these establishments can contribute to the problem by enabling or tolerating behaviors that create opportunities for crime. As such, these spaces can simultaneously function as generators or attractors of crime, or places of informal social control, depending on the local context and

the relationship dynamics between owners, law enforcement, and the communities they serve.

4. Methodology

This section briefly summarizes the hypotheses that guide this research and the methodology used to deepen the connections between the presence of corner stores and other environmental factors with the concentration of three different crime categories (i.e., gun violence, robberies, and assault without firearms) in the City of Newark. With a population of 305,344 as of July 1, 2022 (U.S. Census Bureau, 2024) and an area of approximately 25.88 sq. miles, Newark is New Jersey’s largest city.¹ To account for contextual variation operating across different spatial scales, this study employs a multilevel modeling framework. Specifically, the study area was divided into a regular grid of 200 ft. square cells, which serve as the primary unit of analysis (micro-level), and census block groups, which serve as the secondary unit (meso-level). Additional methodological details regarding the construction of these units and the analytical strategy are provided in the following subsections.

4.1. Hypotheses

The paper starts investigating how different crimes are spatially connected to the presence of corner stores, other convenience stores, and venues sharing similar characteristics (e.g., pharmacies, gas stations, or liquor stores). Based on the literature review and the insights that emerged at the NPSC meetings, the first hypothesis is that corner stores are more likely to concentrate systemic violence than occasional and opportunistic crimes, which, in contrast, are thought to be more prevalent around the other places considered (HP 1).

Second, the study analyzes how these associations are influenced by the characteristics of the surrounding neighborhood. Given the social role of corner stores, it is assumed that their criminogenic influence in the surrounding areas is strictly connected to the socio-economic characteristics of the neighborhoods in which they are located, as stated by the Place in Neighborhood (PIN) theory (Wilcox & Tillyer, 2018). Three specific sub-hypotheses are tested. First, according to the social disorganization theory, a high level of economic hardship should be linked to a higher criminal propensity, thus increasing the general level of crime around corner stores (HP 2.1). Second, a lower level of racial heterogeneity could be linked to a greater centrality of corner stores as social hubs for local communities. This means that people aggregate in and around these places for reasons that go beyond the purchase of food or goods. This may be related to a higher risk of more systemic violence (e.g., linked to gang activities) but also to lower levels of occasional conflicts or opportunistic crimes, due to greater social control (HP 2.2). Third, in areas with a lower access to fresh food, the so-called “food deserts”, a larger percentage of residents likely rely on corner stores for daily purchases, which can increase their exposure to opportunistic crimes (HP 2.3).

Finally, the association between different types of convenience stores and crime can change according to the time of day, thus supporting the identification of distinctive criminogenic mechanisms. Since corner stores are mainly familiar businesses that usually are not open 24/7, it is expected that opportunistic and occasional crimes around them fall during night, while these behaviors may remain concentrated around other places with longer opening hours. On the contrary, more systemic crimes (i.e., gun violence linked to drug dealing) for which corner stores

¹ For the purpose of this analysis, a section of the East Ward of Newark, consisting of the Newark Liberty International Airport and the Port of Newark–Elizabeth Marine Terminal, was excluded due to the industrial nature of this area and the consequent rare occurrence of criminal events within this space

represent a reference place regardless of their actual opening should remain concentrated around corner stores also during nighttime (HP 3).

4.2. Data collection

To assess the different criminogenic mechanisms that could link corner stores with various criminal behaviors, three types of crime were selected. First, following the results of the preliminary analysis and insights from the NCPS, gun violence, including murders and aggravated assaults committed with firearms, was chosen as an indicator of systemic violence, assuming that these events are primarily linked to the activities of local gangs and in connection with drug-dealing activities. Second, aggravated assaults committed without the use of firearms were selected to indicate more impulsive and unplanned violent actions, likely related to personal disputes or alcohol consumption. Finally, robberies were considered to account for a more planned and economically motivated crime, mainly linked to the presence of specific criminal opportunities. Data were provided by the Newark Police Division (NPD) and refer to the geolocation of the single reported events from 1st January to 31st December 2022.

The locations of convenience stores and other relevant businesses were obtained from the company Data-Axle in June 2022. The data file contained information such as business name, primary address, description, and primary SIC code. Any business record listing PO boxes as their main address, as well as non-validated business records, was excluded from the final point-feature vector file. According to Data-Axle, validated business records are verified via direct phone calls to verify data accuracy (i.e., whether the business is open, the primary/mailling address is correct, etc.). Relying on commercial business lists such as Data-Axle (formerly InfoGroup) without validation can be problematic due to issues like incomplete records, business misclassification, or the use of PO boxes as business addresses. For instance, [Lucan et al. \(2013\)](#) evaluated the accuracy of InfoGroup's listings of businesses such as food retailers and restaurants and found that, although suitable for marketing purposes, these data were not reliable for research. Similarly, [Powell et al. \(2011\)](#) reported that, while there was moderate agreement between field observations and commercial business lists for food retailers and restaurants, the accuracy of these lists was significantly lower when assessed based on exact classification matches. Despite these limitations, several strategies can be employed to improve data quality, including reclassifying store types based on business names, verifying location accuracy, and removing duplicate records ([Jones et al., 2017](#)). A comprehensive data cleaning process was implemented to remove duplicated records, incomplete addresses, and listing P.O. boxes as addresses. Lastly, all business records were assessed and verified through direct field observations, with the active participation of community organizers and residents in the process.

Regarding corner stores and convenience stores, all venues were originally embedded within the larger categories of grocery stores or convenience stores under the SIC codes "541,105" and "541,103". To create a specific convenience store category, all business records codified as grocery stores were disaggregated between traditional grocery stores and/or supermarkets (i.e., medium to large food retailers) from small retailers. Furthermore, for the purpose of this study, the determination of what constitutes a corner store was made by visually examining, using Google Street View, each store's business facility to assess their relative size, physical appearance, location, facade condition, and presence of window decals and/or stickers. Of the 126 store businesses, 114 were classified as corner stores, and 12 as other convenience stores, such as corporate-owned or franchise stores. Despite the low number of other convenience stores in Newark, the inclusion of these venues can help identify if corner stores have specific connections to crime compared to other similar activities.

Furthermore, the following environmental features were considered in this study: pharmacies, gas stations, liquor stores, barber shops, retail stores, restaurants, public housing, and vacant lots. Other businesses

were included to provide a more comprehensive picture of local activity patterns and potential attractors for crime that could allow to control for the general environmental context in the area. Pharmacies, gas stations, restaurants, retail stores, and liquor stores were considered because of their proximity to convenience stores in terms of business activities and potential connections with criminal opportunities. All are indeed cash-intensive activities, open to the public, selling a wide range of products, and with long operational hours ([Bernasco & Block, 2011](#); [La Vigne & Wartell, 2015](#); [Taylor, 2002](#); [Toomey et al., 2012](#)). In contrast, barber shops were included since they can be targets for crime, such as robberies, but also play a role as social hubs, especially in urban communities with minority populations of Blacks and Latinos, such as Newark, acting as protective factors against crime ([Boehme, 2023](#)). Business records for pharmacies ($n = 72$), gas stations ($n = 30$), liquor stores ($n = 55$), restaurants ($n = 412$), retail stores ($n = 125$), and barber shops ($n = 30$) within the boundaries of the City of Newark were obtained from the company Data-Axle.

The study also includes the presence of public housing real estate and vacant lots. Public housing is often associated with crime due to the concentration of economic and social disadvantages and poor building design ([Haberman et al., 2013](#); [Holzman et al., 2005](#); [Kelsay & Haberman, 2021](#); [Weatherburn et al., 1999](#)). Data were obtained from the Newark Housing Authority, which comprises 42 public housing development complexes located throughout the city, with nearly 45 % of them in the Central Ward and 33 % in the South Ward. All public housing units were analyzed as polygon-feature vector files. Vacant lots have often been associated with crime and deviant activities (i.e., drug consumption or prostitution) due to lower formal and informal control ([Boessen & Chamberlain, 2017](#); [Chen & Rafail, 2020](#)). In Newark, vacant lots are often located in disadvantaged neighborhoods and help shape the spatial patterns of crime by influencing social interaction and informal surveillance. When neglected, vacant lots can reinforce perceptions of disorder and weaken informal social control, creating conditions that are more conducive to illicit activities such as drug use and sales. Data on vacant lots were retrieved from official municipal records readily accessible on the City of Newark's Open Data portal.

Finally, this study also considers the conditions of the different Newark neighborhoods to assess their potential influence on the association between crime and corner stores. This is in line with the PIN theory, assuming that neighborhood characteristics can influence the role of environmental factors at the micro-level in producing crime-inducing environments ([Wilcox & Tillyer, 2018](#)). Three variables were selected. First, an index of the level of economic hardship was created applying a factor analysis with principal component factor extraction method and orthogonal varimax rotation of the obtained latent factors to four socio-economic variables, namely the unemployment rate, the poverty rate, the percentage of single mother households, and the cost of living. The factor analysis highlights one significant principal factor significantly correlated with all the four original variables that was interpreted as an economic hardship index. Second, a racial heterogeneity score was calculated for each neighborhood considering the shares of the resident population by ethnicity.² The lower the score, the more residents of a neighborhood belong to the same ethnicity. All these indicators were calculated relying on data obtained from the U.S. Census Bureau American Community Survey (ACS) estimates for the year 2022. Finally, a low accessibility score for fresh food was included. The score was compiled by the New Jersey Economic Development Authority (NJEDA) and indicates the degree to which residents are underserved by supermarkets in their neighborhood, and it ranges from 0 to 100 for each U.S. Census block group within the City of Newark. Residents of a block

² The score was calculated using the following formula: Racial heterogeneity score = $1 - \sum p_i^2$. Where p_i represents the percentages of total resident population categorized respectively as non-Hispanic white, African-American or Hispanic persons ([Kader & Perry, 2007](#)).

group with a higher Low Access Score typically travel longer distances to access a major supermarket than residents of a block group with a lower Low Access Score. Higher values of the score can indicate that residents must rely more on corner stores or convenience stores to access food and other products.

4.3. Analytical strategy

This study investigates the relationship between the three selected crimes, the presence of corner stores, and other environmental factors using a set of multilevel Poisson regression models, including some interaction effects. The decision to employ a Poisson functional form is driven by the countable nature of the outcome variables (i.e., the number of events). Additionally, the distributions of the outcome variables do not show signs of overdispersion. The choice of using a multilevel model is based on both theoretical and statistical motivations. On the one hand, this model allows for consideration of the possibility that a criminal event may be associated with environmental or societal factors operating at different geographical levels (Dugato, 2022). On the other hand, by acknowledging the nested structure of the data, multilevel models address the correlation between residuals arising from the grouping of units of analysis into distinct clusters. This approach considers associations within and between units of analysis at higher levels, effectively controlling for potential biases in the estimation process and enhancing the reliability of estimates (Johnson, 2010).³

In particular, the area of the City of Newark, NJ, was subdivided into a regular grid of 11,467 square cells, each with a side length of 200 ft. These cells estimate a half of the size of a block and represent the first level of analysis (micro-level).⁴ The dependent variables are the count of recorded crime events in 2022 within each cell for gun violence, robbery and aggravated assault without firearms, respectively. To account for the potential temporal variation in the analyzed associations, two additional variables were generated for each crime category distinguishing the count of events recorded during daytime (from 7 am to 6 pm) or night-time (from 7 pm to 6 am). Regarding the spatial influence of corner stores and other environmental factors at the micro-level, a set of dummy variables was created for proximity within 200 ft from corner stores, other convenience stores, pharmacies, gas stations, liquor stores, barber shops, restaurants, retail stores, public housing, and vacant lots. In addition, for corner stores and other convenience stores only, two other dummy variables are defined identifying cells between 200 and 800 ft from these venues. These latter variables allow us to estimate whether crime events concentrate around these places, also considering a wider radius.

Moving to the *meso* level of analysis, 201 census block groups were considered for approximating between-neighborhood variations. These block groups were characterized by their corresponding values of the low access score to fresh food, the economic hardship index, and the racial heterogeneity score. All three variables were standardized using a z-score linear transformation to facilitate comparability of the coefficients resulting from the regression models. Each cell within the regular grid was associated with the census block groups containing its centroid, establishing a perfectly nested two-level system. The descriptive statistics for the defined variables at both the grid cell and census

block group levels are summarized in Table 1. Furthermore, to assess the variation of associations between proximity to corner stores and crime conditional to neighborhood characteristics, a set of interaction terms were also included in the regression models and the corresponding marginal predicted means were calculated for facilitating their interpretation.

5. Results

This section presents the results of the regression analysis performed to understand the spatial dynamics between corner stores, other convenience stores, and various types of crime. The first three regression models aim to test the first hypothesis, posing that corner stores are more likely to concentrate systemic violence compared to occasional or opportunistic crimes, which are expected to be more prevalent around other types of commercial or public spaces. This hypothesis is partially supported by the data (Table 2). Corner stores are significantly associated with all three categories of crime considered in the analysis: systemic, opportunistic, and occasional. This broad association reflects the multifunctional nature of corner stores, which may act not only as commercial outlets but also as informal social hubs, potentially concentrating various deviant or criminal behaviors. Specifically, corner stores are located in areas with high levels of gun violence events, whereas this crime seems not to concentrate significantly around similar businesses, such as convenience stores or pharmacies. The other businesses significantly associated with gun violence are liquor stores, barber shops and restaurants, which mainly include fast foods or quick service restaurant. These places may similarly favor social gathering around the premises.

Additionally, the magnitude of the coefficients suggests that the likelihood of observing a gun violence event in cells within 200 ft from a corner store is appreciably greater than those recorded for aggravated assaults without firearms. This crime appears more strongly associated with public housing and liquor stores, indicating possible dynamics related to substance-consumption that may be a key driver of spontaneous conflicts, or public housing, which may associate with cases of

Table 1
- Descriptive statistics of the variables used

Variable	N	Mean	Std. Dev.	Min	Max
Crime					
Gun Violence	11,467	0.03	0.18	0.00	5.00
Gun Violence (day)	11,467	0.01	0.12	0.00	2.00
Gun Violence (night)	11,467	0.02	0.13	0.00	5.00
Agg Assault (no gun)	11,467	0.05	0.25	0.00	5.00
Agg Assault (no gun) (day)	11,467	0.02	0.15	0.00	2.00
Agg Assault (no gun) (night)	11,467	0.03	0.19	0.00	5.00
Robbery	11,467	0.04	0.24	0.00	6.00
Robbery (day)	11,467	0.02	0.14	0.00	4.00
Robbery (night)	11,467	0.03	0.18	0.00	5.00
Micro-level factors					
Corner stores – 200 ft	11,467	0.08	0.26	0.00	1.00
Corner stores – 800 ft	11,467	0.43	0.50	0.00	1.00
Other Convenience stores – 200 ft	11,467	0.01	0.09	0.00	1.00
Other Convenience stores – 800 ft	11,467	0.06	0.23	0.00	1.00
Pharmacies	11,467	0.05	0.21	0.00	1.00
Gas Stations	11,467	0.02	0.13	0.00	1.00
Liquor Stores	11,467	0.04	0.19	0.00	1.00
Barber Shops	11,467	0.02	0.14	0.00	1.00
Retail Stores	11,467	0.05	0.23	0.00	1.00
Restaurants	11,467	0.16	0.37	0.00	1.00
Public Housing	11,467	0.12	0.33	0.00	1.00
Vacant Lots	11,467	0.16	0.37	0.00	1.00
Meso-level factors					
Low Access to Fresh Food	201	0.00	1.00	-0.84	1.89
Economic hardship	201	0.00	1.00	-2.16	4.75
Racial Heterogeneity	201	0.00	1.00	-1.75	2.39

³ The suitability of a multilevel random intercept model versus a fixed effect model was assessed for each specification using a Hausman test.

⁴ The choice of the most suitable spatial unit of analysis remains a debated issue in geographic criminology (Hipp, 2007; Tita & Radil, 2010; Weisburd et al., 2009). In this study we decided to opt for a regular grid instead of street segments, considering that in Newark, street segments vary considerably in length and shape and often intersect at locations where corner stores are concentrated. Using a regular grid of square cells provides a more standardized spatial framework that enables consistent comparisons and allows the analysis to better represent localized spatial influence patterns at the micro-level.

Table 2

Incidence rate ratios estimated from Poisson regression models assessing the relationship between the presence of a corner store and other contextual variables with three types of crime outcomes with robust standard errors. 95 % confidence intervals are shown in brackets

	Gun Violence		Aggravated Assault (no firearms)		Robbery	
	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)
Micro-level factors						
Corner stores – 200 ft	2.50***	[1.63–3.85]	1.45*	[1.02–2.05]	1.79**	[1.24–2.58]
Corner stores – 800 ft	1.45*	[1.04–2.03]	1.17	[0.92–1.50]	1.43**	[1.10–1.87]
Other Conv. stores – 200 ft	2.24	[0.68–7.45]	0.83	[0.28–2.41]	2.33*	[1.07–5.09]
Other Conv. stores – 800 ft	0.84	[0.41–1.73]	0.70	[0.46–1.06]	1.18	[0.72–1.91]
Pharmacies	0.47*	[0.23–0.96]	1.25	[0.81–1.94]	0.71	[0.44–1.14]
Gas Stations	0.98	[0.39–2.51]	1.48	[0.88–2.47]	3.32***	[2.27–4.86]
Liquor Stores	1.63*	[1.05–2.52]	2.06***	[1.49–2.86]	1.76**	[1.20–2.58]
Barber Shops	2.08*	[1.12–3.86]	1.45	[0.90–2.34]	2.61***	[1.60–4.27]
Retail Stores	1.25	[0.77–2.04]	1.89***	[1.30–2.76]	1.41*	[1.01–1.97]
Restaurants	1.61**	[1.15–2.25]	1.46*	[1.09–1.95]	2.13***	[1.59–2.85]
Public Housing	1.34	[0.86–2.09]	2.03***	[1.52–2.70]	1.11	[0.80–1.55]
Vacant Lots	1.59**	[1.19–2.12]	1.35*	[1.05–1.73]	1.22	[0.90–1.67]
Meso-level factors						
Racial Heterogeneity	0.72***	[0.61–0.86]	1.02	[0.91–1.15]	0.97	[0.86–1.08]
Economic hardship	1.31***	[1.11–1.54]	1.09	[0.94–1.26]	0.98	[0.85–1.14]
Low Access to Fresh Food	1.24**	[1.06–1.43]	1.06	[0.93–1.21]	1.01	[0.89–1.14]
Constant	0.01***	[0.01–0.02]	0.03***	[0.02–0.03]	0.02***	[0.02–0.03]
var(Constant)	0.54***	[0.27–1.07]	0.42***	[0.27–0.63]	0.26***	[0.16–0.42]

* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

interpersonal violence against acquaintances or neighbors. Relevant is also the association with retail stores, indicating concentrations of occasional violence in commercial areas probably due to flows of people.

Also significant is the coefficient connecting the likelihood of robberies and the proximity of corner stores. However, the alleged criminogenic effect of corner stores is lower than that recorded by other cash-intensive activities, such as gas stations, barber shops, restaurants, and other convenience stores. This suggests that all these businesses primarily attract crimes due to their shared characteristics that generate criminal opportunities making them or their clients high-yield targets,

rather than for specific social or cultural mechanisms. As an additional consideration, it is also interesting that proximity to barber shops is linked to significantly higher concentrations of both gun violence and robberies, partially contrasting their supposed role of protective factors from crimes (e.g. [Boehme, 2023](#)).

Finally, gun violence is the only crime type significantly linked to more structural socio-economic marginality and demographic composition, as evidenced by its spatial connections with disadvantaged neighborhood characteristics. This may support the assumption that systemic violence is the consequence of social processes that are more

Table 3

- Incidence rate ratios estimated from Poisson regression models assessing the relationship between the presence of corner stores and other contextual variables with three types of crime outcomes including interaction terms between corner stores presence and neighborhood characteristics with robust standard errors. 95 % confidence intervals are shown in brackets

	Gun Violence		Aggravated Assault (no firearms)		Robbery	
	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)
Micro-level factors						
Corner stores – 200 ft	2.53***	[1.58–4.04]	1.54**	[1.11–2.14]	1.71**	[1.19–2.46]
Corner stores – 800 ft	1.47*	[1.03–2.10]	1.15	[0.90–1.46]	1.35*	[1.04–1.75]
Other Conv. stores – 200 ft	2.14	[0.59–7.69]	0.86	[0.29–2.53]	2.45*	[1.15–5.21]
Other Conv. stores – 800 ft	0.82	[0.40–1.68]	0.70	[0.46–1.06]	1.20	[0.75–1.90]
Pharmacies	0.46*	[0.22–0.94]	1.23	[0.79–1.91]	0.69	[0.43–1.12]
Gas Stations	0.98	[0.39–2.50]	1.50	[0.90–2.49]	3.41***	[2.34–4.96]
Liquor Stores	1.63*	[1.05–2.51]	2.11***	[1.51–2.93]	1.81**	[1.23–2.66]
Barber Shops	2.09*	[1.13–3.86]	1.49	[0.92–2.42]	2.75***	[1.68–4.50]
Retail Stores	1.25	[0.77–2.04]	1.85***	[1.27–2.70]	1.39*	[1.00–1.95]
Restaurants	1.61**	[1.15–2.25]	1.46**	[1.10–1.95]	2.10***	[1.57–2.82]
Public Housing	1.34	[0.86–2.09]	2.09***	[1.57–2.78]	1.14	[0.82–1.59]
Vacant Lots	1.59***	[1.20–2.11]	1.36*	[1.06–1.73]	1.26	[0.93–1.71]
Meso-level factors						
Racial Heterogeneity	0.70*	[0.54–0.92]	1.03	[0.87–1.24]	0.82*	[0.70–0.97]
Economic hardship	1.29	[0.97–1.72]	1.13	[0.89–1.44]	0.90	[0.71–1.14]
Low Access to Fresh Food	1.29*	[1.00–1.66]	1.10	[0.91–1.34]	1.11	[0.94–1.32]
Interactions						
Corner stores – 200 ft. X Racial Heterogeneity	0.99	[0.68–1.46]	0.92	[0.66–1.26]	1.43*	[1.06–1.94]
Corner stores – 200 ft. X Economic hardship	1.06	[0.73–1.55]	0.77	[0.51–1.17]	0.97	[0.63–1.50]
Corner stores – 200 ft. X Low Access to Fresh Food	0.88	[0.60–1.28]	0.82	[0.62–1.09]	0.86	[0.61–1.20]
Corner stores – 800 ft. X Racial Heterogeneity	1.06	[0.74–1.52]	1.02	[0.81–1.27]	1.28	[0.98–1.67]
Corner stores – 800 ft. X Economic hardship	1.01	[0.72–1.40]	0.98	[0.73–1.30]	1.23	[0.93–1.64]
Corner stores – 800 ft. X Low Access to Fresh Food	0.96	[0.71–1.30]	0.98	[0.79–1.23]	0.83	[0.65–1.05]
Constant	0.01***	[0.01–0.02]	0.03***	[0.02–0.03]	0.02***	[0.02–0.03]
var(Constant)	0.53	[0.27–1.06]	0.43	[0.28–0.65]	0.22	[0.13–0.38]

* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

likely to emerge in deprived contexts.

This latter point drives the second hypothesis related to the connection between the criminogenic role of corner stores and the surrounding environment. According to the PIN theory, the relationship between corner stores and crime is expected to be mediated or intensified by the surrounding social and economic context, given the role of corner stores as social aggregation points. Specifically, three sub-hypotheses were formulated and tested by introducing in the three regression models interaction effects between variables representing spatial proximity to corner stores and neighborhood-level variables (Table 3). To facilitate interpretation and assess the relevance of these findings for the proposed hypotheses, marginal predicted means were calculated on the basis of the model results. Indeed, previous research has shown that in nonlinear and multilevel models, the statistical significance of an interaction coefficient does not always reflect the magnitude or substantive relevance of the underlying effect (Brambor et al., 2006; Mize, 2019; Mustillo et al., 2018). Therefore, we calculate the marginal predicted means that represent the expected average number of criminal events in cells located within or beyond a specified distance from a corner store, 200 ft or 800 ft, calculated at representative values of a given neighborhood covariate (Williams, 2012). This approach can reveal meaningful differences across combinations of environmental characteristics even when the overall interaction coefficients are not statistically significant.

The first sub-hypothesis (2.1) proposed that lower access to fresh food would increase the reliance on corner stores for food provisioning by local residents, thus heightening their exposure to opportunistic crimes such as robbery. This hypothesis is not verified. In contrast, the marginal predicted mean of the number of robberies estimated in cells close to a corner store at different values of low access to fresh food indicates that corner stores are significantly associated with a higher risk of robberies only in areas with higher access to fresh food. For example, in neighborhoods with high accessibility to fresh food (that is, when the corresponding index is -0.5 standard deviation from the mean), the expected value for robberies in cells within 200 ft from a corner store is more than double the predicted value for other cells. However, when the level of low access to fresh food is above the mean, the expected value of robberies for cells located within 200 ft of corner stores is no longer statistically different than the one predicted for the other cells in the dataset (Fig. 1). The same dynamics also emerge considering a radius of 800 ft around corner stores. One possible explanation is that in areas where corner stores are not the primary source of food, they are less integrated into the social fabric, and thus less frequented and more vulnerable to crime due to lower informal control.

The second sub-hypothesis (2.2) suggested that in more racially homogeneous neighborhoods, corner stores function as social anchors or gathering points which may increase systemic violence while reducing

occasional interpersonal conflicts and opportunistic crimes. This hypothesis is supported by the results. In fact, looking at the marginal predicted means, racial heterogeneity partially offsets the criminogenic influence of corner stores for gun violence, but only when heterogeneity exceeds a threshold approximately 0.5 standard deviations above the mean. This suggests that corner stores significantly increase the likelihood of gun violence only in relatively homogeneous neighborhoods, as hypothesized (Fig. 2).

In contrast, for robberies, a different pattern emerges: the interaction between corner stores and racial heterogeneity is positive and significant considering both a 200- and 800-ft radius, implying that corner stores located in more racial diverse neighborhoods are significantly more likely to concentrate a higher number of robberies around them. In other areas, in contrast, increased racial heterogeneity does not significantly affect the likelihood of robbery (Fig. 3).

The third sub-hypothesis (2.3) posited that higher levels of economic hardship would correspond with elevated criminal risk around corner stores. This hypothesis is partially supported. Although economic hardship is directly correlated with a higher likelihood of gun violence, it does not substantially amplify the criminogenic role of corner stores (Fig. 4). Specifically, the ratio between the predicted number of gun violence events in cells close to corner stores and the one expected in other areas is 2.32 in neighborhoods with very low economic hardship (-1 sd) and 2.63 in areas facing greater economic strain ($+1$ sd).

Finally, hypothesis 3 focused on whether the relationship between different types of convenience-oriented businesses and crime varied by time of day. It was assumed that, since corner stores are familiar and locally embedded businesses that typically do not operate 24/7, they would be more vulnerable to opportunistic and occasional crimes during the day, while more systemic or structural violence (e.g., linked to drug markets) might persist in nighttime hours. To test this hypothesis, the three models were replicated by dividing crime events that occurred during daytime hours (7 am – 6 pm) from those recorded during nighttime hours (7 pm – 6 am). As shown in Table 4, the results allow us to only partially verify the hypothesis.

In general, corner stores exhibit significant association with crime both during day and night hours. This suggests that the criminal activity clustered around these venues is not solely tied to their business operations but may also reflect informal or illicit activity in their vicinity, such as drug transactions. Regarding other venues similar to corner stores, restaurants emerge as significant hotspots for gun violence and aggravated assaults during the night, consistent with their frequent long operational hours. In contrast, other convenience stores show a significant relationship with robberies only during the day. This may seem contradictory, given that many branded stores are open overnight. However, a closer look reveals that many of these stores are located near gas stations, which exhibit the strongest association with nighttime

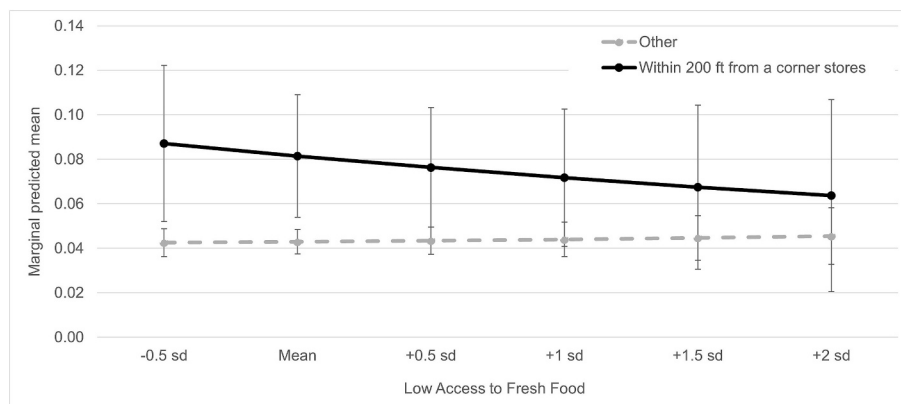


Fig. 1. Marginal predicted mean of the number of robberies by different values of the Low Access to Fresh Food index, comparing cells within 200 ft of a corner store and other cells (with 95 % CIs).

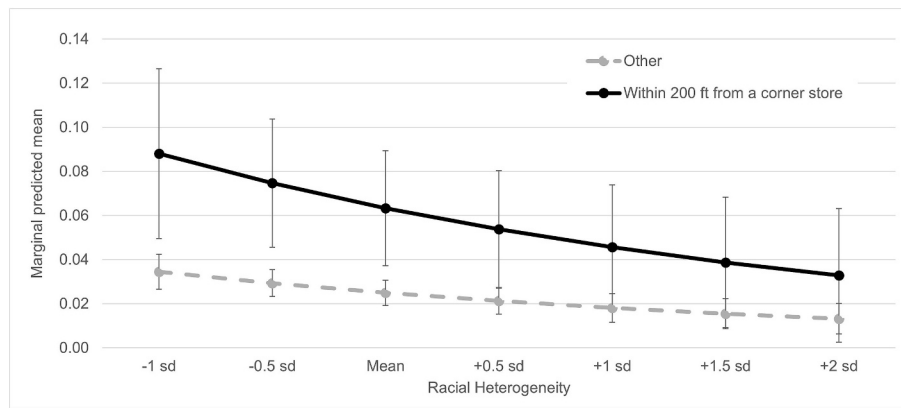


Fig. 2. Marginal predicted mean number of gun violence events by different values of racial heterogeneity, comparing cells within 200 ft of a corner store and other cells (with 95 % CIs).

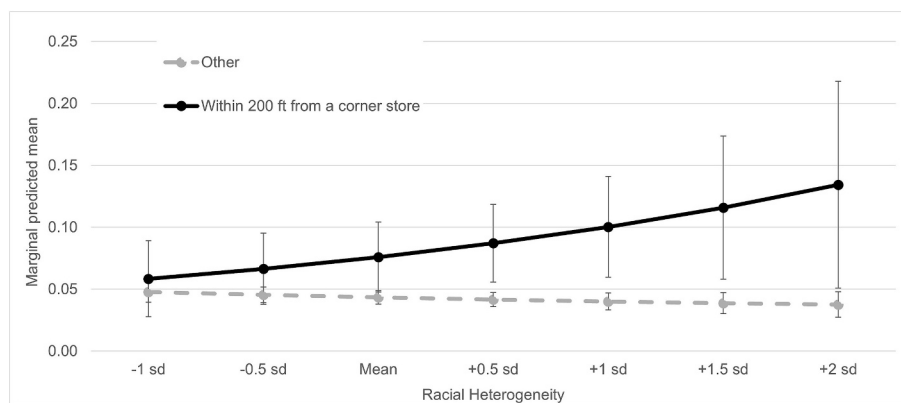


Fig. 3. Marginal predicted mean number of robberies by different values of racial heterogeneity, comparing cells within 200 ft of a corner store and other cells (with 95 % CIs).

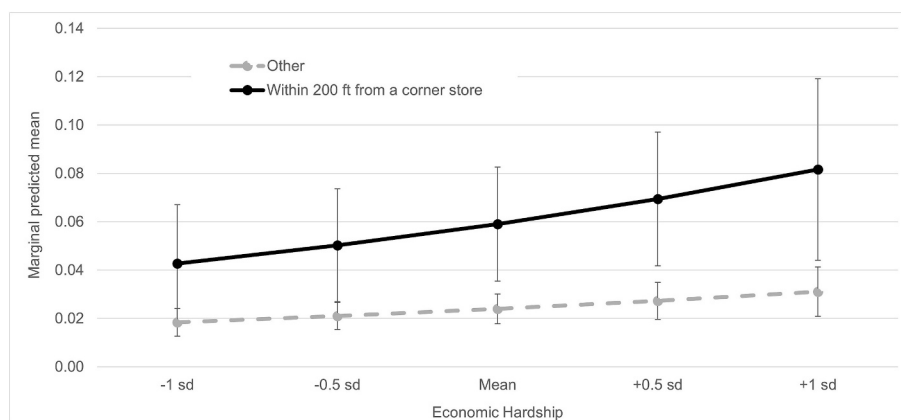


Fig. 4. Marginal predicted mean number of gun violence events by different values of economic hardship, comparing cells within 200 ft of a corner store and other cells (with 95 % CIs).

robberies. This spatial co-location could explain the differential temporal risk profiles. These temporal distinctions underscore the importance of disaggregating crime data by time to fully capture the risk dynamics associated with different types of business.

6. Discussion and conclusions

The findings of this study have both theoretical and practical implications, shedding light on the relationship between crime and specific

features of the built environment, with a particular focus on corner stores and similar establishments. In general, these results not only support the previous intuition that corner stores are not equal to other similar businesses or venues but also suggest that there are significant differences among corner stores themselves.

On the one hand, although corner stores are largely present in the study area, all three types of crime are significantly more concentrated around these locations. This is evidenced by the fact that, in all models, crime events are significantly more likely to occur in the cells within

Table 4

Incidence rate ratios estimated from Poisson regression models assessing the relationship between the presence of corner stores and other contextual variables with three types of crime outcomes and different times of the day including interaction terms between corner stores presence and neighborhood characteristics with robust standard errors. The 95 % confidence intervals are shown in brackets

	Gun Violence				Aggravated Assault (no firearms)				Robbery			
	Day		Night		Day		Night		Day		Night	
	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)	IRR	CI (95 %)
Micro-level factors												
Corner stores – 200 ft	2.30*	[1.19–4.44]	3.09***	[1.72–5.54]	1.68*	[1.10–2.57]	1.47	[0.97–2.23]	1.68	[0.99–2.88]	1.69*	[1.08–2.66]
Corner stores – 800 ft	1.34	[0.80–2.25]	1.77*	[1.11–2.83]	1.41*	[1.04–1.93]	1.05	[0.77–1.43]	1.50*	[1.05–2.14]	1.25	[0.92–1.70]
Other Convenience stores – 200 ft	2.81	[0.55–14.49]	1.45	[0.23–9.09]	0.28	[0.03–2.21]	1.50	[0.51–4.37]	3.75**	[1.43–9.82]	1.83	[0.61–5.48]
Other Convenience stores – 800 ft	0.85	[0.34–2.11]	0.75	[0.34–1.64]	1.06	[0.52–2.16]	0.49	[0.24–1.02]	1.19	[0.53–2.66]	1.20	[0.77–1.86]
Pharmacies	0.21*	[0.06–0.77]	0.67	[0.30–1.54]	1.14	[0.67–1.91]	1.35	[0.78–2.34]	0.64	[0.31–1.32]	0.75	[0.45–1.25]
Gas Stations	0.46	[0.06–3.60]	1.38	[0.41–4.60]	1.81	[0.88–3.73]	1.28	[0.74–2.20]	1.90	[0.91–3.94]	4.66***	[3.18–6.83]
Liquor Stores	2.08**	[1.21–3.58]	1.20	[0.61–2.33]	2.23***	[1.49–3.34]	2.05***	[1.34–3.14]	2.01*	[1.16–3.47]	1.67*	[1.07–2.61]
Barber Shops	3.00**	[1.42–6.34]	1.34	[0.61–2.96]	1.13	[0.62–2.07]	1.68	[0.83–3.39]	2.72*	[1.26–5.91]	2.67***	[1.65–4.32]
Retail Stores	1.46	[0.65–3.28]	1.03	[0.52–2.04]	1.37	[0.83–2.26]	2.18***	[1.39–3.41]	1.69*	[1.05–2.72]	1.24	[0.82–1.88]
Restaurants	1.34	[0.83–2.17]	1.79**	[1.17–2.72]	1.38	[0.91–2.10]	1.60**	[1.12–2.28]	2.14***	[1.50–3.05]	2.12***	[1.48–3.03]
Public Housing	1.44	[0.82–2.52]	1.27	[0.77–2.09]	1.97***	[1.42–2.75]	2.09***	[1.49–2.95]	1.40	[0.84–2.34]	1.02	[0.72–1.45]
Vacant Lots	2.09**	[1.30–3.37]	1.41	[0.97–2.05]	1.21	[0.85–1.73]	1.53*	[1.10–2.12]	1.25	[0.80–1.95]	1.31	[0.93–1.85]
Meso-level factors												
Racial Heterogeneity	0.67*	[0.47–0.96]	0.74	[0.52–1.05]	0.93	[0.72–1.21]	1.09	[0.88–1.36]	0.67**	[0.51–0.89]	0.91	[0.75–1.11]
Economic hardship	1.08	[0.82–1.43]	1.36	[0.94–1.95]	1.17	[0.87–1.57]	1.07	[0.81–1.40]	0.89	[0.67–1.20]	0.87	[0.66–1.14]
Low Access to Fresh Food	1.15	[0.81–1.61]	1.37*	[1.01–1.86]	1.26	[0.98–1.61]	1.01	[0.81–1.26]	1.09	[0.84–1.43]	1.14	[0.93–1.38]
Interactions												
Corner stores – 200 ft. X Racial Heterogeneity	1.29	[0.80–2.09]	0.87	[0.53–1.44]	0.95	[0.63–1.44]	0.94	[0.62–1.42]	1.61	[0.99–2.61]	1.37	[0.98–1.91]
Corner stores – 200 ft. X Economic hardship	1.28	[0.75–2.19]	0.98	[0.66–1.46]	0.95	[0.58–1.55]	0.72	[0.43–1.20]	1.15	[0.66–1.99]	0.93	[0.52–1.64]
Corner stores – 200 ft. X Low Access to Fresh Food	0.91	[0.56–1.48]	0.83	[0.53–1.28]	0.79	[0.55–1.12]	0.81	[0.55–1.20]	0.63	[0.36–1.12]	1.00	[0.68–1.47]
Corner stores – 800 ft. X Racial Heterogeneity	1.20	[0.74–1.95]	1.03	[0.65–1.65]	1.12	[0.83–1.52]	0.96	[0.71–1.29]	1.19	[0.81–1.74]	1.39*	[1.01–1.91]
Corner stores – 800 ft. X Economic hardship	1.26	[0.88–1.81]	0.90	[0.60–1.34]	0.93	[0.64–1.35]	1.10	[0.80–1.52]	1.00	[0.68–1.49]	1.46*	[1.04–2.04]
Corner stores – 800 ft. X Low Access to Fresh Food	0.97	[0.66–1.43]	0.92	[0.62–1.37]	0.82	[0.63–1.07]	1.11	[0.82–1.52]	0.93	[0.66–1.30]	0.74	[0.55–1.01]
Constant	0.01***	[0.00–0.01]	0.01***	[0.00–0.01]	0.01***	[0.01–0.02]	0.01***	[0.01–0.02]	0.01***	[0.01–0.01]	0.01***	[0.01–0.02]
var(Constant)	0.41	[0.14–1.14]	0.49	[0.22–1.12]	0.38	[0.21–0.67]	0.48	[0.30–0.77]	0.39	[0.21–0.75]	0.20	[0.09–0.44]

* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

200 ft of these venues than in the subsequent areas from 200 ft to 800 ft away. On the other hand, it must be noted that the three types of crime considered are concentrated in different areas of the city of Newark. This entails that, although on average corner stores are associated with a higher level of all three crimes across the whole city, in reality only a few of them are located within the main concentrations for each crime, and the corner stores associated with higher levels of one crime are not likely the same linked with other illicit behaviors.

For example, only 47 of the 114 corner stores (42.1 %) are located within 200 ft of at least one of the 1 % of the cells with the highest predicted values for each type of crime. However, only three of them are associated with high risk for all three crimes, and the majority (32–68 %) of the corner stores associated with the highest concentrations are linked only to one specific crime risk, especially gun violence (Fig. 5). These differences may be due to the specificities of the single corner store, but also to the combined influence of other environmental factors that may affect the different activities that occur around these locations, eventually attracting or generating different criminal behaviors. This highlights the necessity to move beyond generic risk classifications toward more place-specific diagnostics.

Specifically, by looking at the other environmental factors more commonly associated with corner stores in these high-risk areas, it is possible to underscore how different types of offenses exhibit distinct relationships with the built environment and derive the potential role of corner stores in influencing crime concentrations. Aggravated assaults, used here as a proxy for occasional and unplanned violence, tend to concentrate primarily in areas associated with alcohol consumption, drug misuse, informal social gatherings and concentrations of economic disadvantages and racial heterogeneity. In these contexts, corner stores may function mainly as crime generators, facilitating unstructured social interactions and congregation, often under conditions of limited formal or informal supervision. In contrast, robberies, understood as more premeditated and economically motivated offenses, cluster not only around corner stores but also around other types of cash-intensive small businesses, such as gas stations, restaurants, retail stores and barber shops. Further, socio-economic hardship appears to be less strongly associated with the concentration of robberies around these

corner stores. In both cases, aggravated assaults and robberies, the opportunity structure framework proves valuable: crimes appear to occur where situational conditions, due to the presence of specific business or economic activities, are most conducive to their commission.

Gun violence, however, seems to follow a different pattern. The presence of corner stores in high-risk areas for this crime is often accompanied by the proximity to vacant lots and public housing. Given that gun violence is conceptualized here as less opportunistic and more embedded in systemic or drug-related dynamics, these findings align with narratives collected during NPSC community meetings, in which corner stores were described as local hubs not only for legitimate social activity, but also for illicit interactions, including drug dealing. This supports previous findings by Barnum et al. (2017), who observed that drug dealers tend to cluster around locations with high visibility and recognizability from potential clients and low supervision. This interpretation is also confirmed by the fact that the corner stores at higher risk for gun violence are also located in areas with low access to fresh food, which increases the centrality of these businesses for the local communities.

However, the fact that corner stores, but not other comparable venues such as pharmacies or other convenience stores, are significantly associated with gun violence suggests also that their criminogenic role cannot be attributed solely to their business function (i.e., the occasional convergence of many people). Instead, it indicates that corner stores can play unique social and environmental roles within the community that make them more exposed to criminal conduct, eventually leading to episodes of gun violence. Further, the results of the analysis show that corner stores remain associated with crime across both daytime and nighttime hours, suggesting that their role extends beyond commercial activity to encompass informal social dynamics and, potentially, more systemic criminal processes such as drug dealing.

This conclusion is also supported by observing that the other places significantly associated with higher gun violence in their proximity are liquor stores and barber shops, both of which are characterized by being places of social gatherings and interactions similar to corner stores. For example, local community accounts support the intuition that some barber shops may operate as informal social hubs, where the

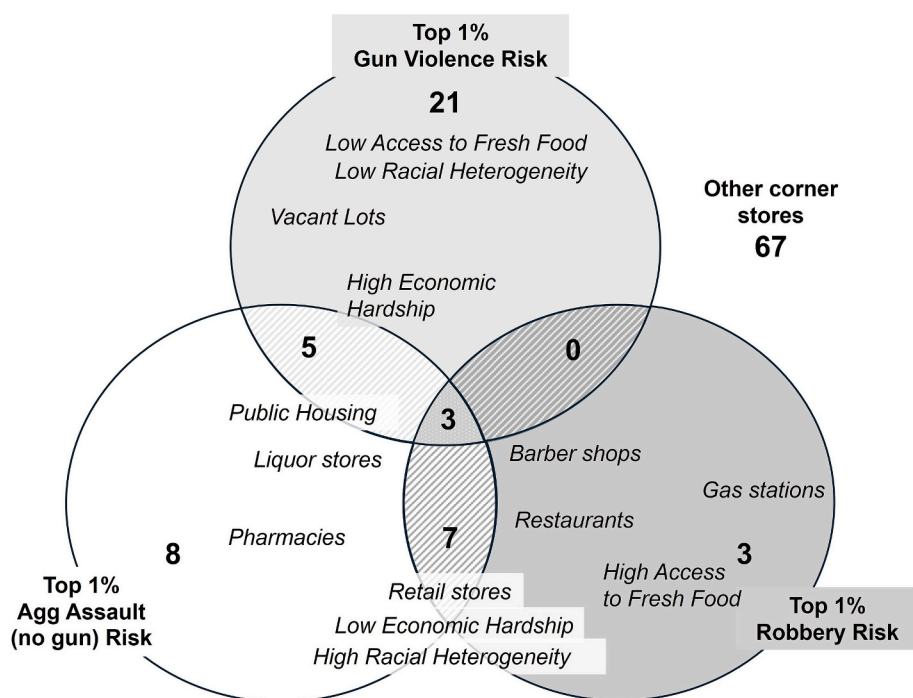


Fig. 5. Number of corner stores within 200 ft of the 1 % cells with the highest predicted number of crime events by type of crime and including the other environmental factors more commonly present in the surrounding areas.

unauthorized sale and consumption of alcohol, illegal gambling activities, and, in some cases, the distribution of illicit substances occur. These practices may amplify the criminogenic potential of such establishments, as the surrounding areas become more vulnerable due to the convergence of intoxicated individuals, high target density, and limited guardianship.

These findings, especially the comparison between corner stores and branded convenience stores, are quite relevant since they partially contrast the assumption that locally owned small businesses are more likely to exert a protective action on criminal behaviors than activities owned by non-local residents (Kim & Hipp, 2022; Linning & Eck, 2021). However, it should be acknowledged that the focus of these previous studies is slightly different in terms of types of businesses and types of crime considered. For example, our results also show that robberies are more concentrated around branded convenience stores than corner stores, thus supporting these previous studies. In this sense, our study may represent an exception to a wider trend given by the specific social role of corner stores in the examined areas. Further, although we can reasonably assume that corner stores are more likely to be locally owned than branded convenience stores, we do not have precise information to test this assumption. Therefore, this conclusion should be taken with caution.

Moreover, the multilevel models further reveal that neighborhood-level structural characteristics are significantly associated with both gun violence and robbery, indicating that the concentrations of these crimes are shaped not only by opportunity structures at the micro level but also by broader socio-structural factors. This complicates conventional environmental criminology models that emphasize place-based opportunity and instead highlights the need to integrate *meso*-level neighborhood characteristics when interpreting crime concentrations at micro-places. Moreover, significant interaction effects between closeness to corner stores and neighborhood characteristics in determining the expected number of events for both crimes suggest a dynamic relationship between micro- and meso-level factors. This supports two complementary interpretations, consistent with the PIN theory.

First, neighborhood-level socioeconomic and demographic traits may influence how micro-places are used and perceived by residents, which in turn affects the degree of deviant behavior concentrated there. For example, the finding that robbery concentrates more around corner stores in neighborhoods with greater access to fresh food may reflect their different social functions in such contexts. In these areas, corner stores may not serve as primary providers of food or other basic goods, since residents have access to more varied and often expensive alternatives. Consequently, this is likely to change the profile of the corner store clientele. This may increasingly include economically marginalized individuals or groups using these spaces for non-commercial purposes, including informal congregation. On the contrary, in neighborhoods with lower food access, corner stores serve a broader and more diverse customer base, potentially increasing natural surveillance and informal guardianship in the place and the surrounding area. Second, an alternative reading of the micro-meso relationship is also plausible: certain micro-places may possess sufficient ecological or social distinctiveness to modify the overall influence of neighborhood-level characteristics on crime. Rather than neighborhood conditions simply mediating the criminogenic potential of places, places themselves may amplify or attenuate broader structural effects. In summary, these findings highlight the multifaceted role that different types of retail and commercial spaces play in shaping urban crime patterns. Additionally, they underscore the importance of accounting for the spatial and social context in which these places are embedded.

From a policy perspective, the results have several implications for urban crime prevention and spatial planning. Primarily, they support the idea that place-based interventions should be differentiated by crime type and informed by context-specific environmental diagnostics (Kennedy et al., 2018). In general, this study supports the conclusion that to reduce crime around corner stores in high-risk areas these places

should be evaluated not only for their business practices but also for their embeddedness in wider social and physical contexts. For example, for gun violence, traditional opportunity reduction measures may be insufficient. Instead, effective prevention must combine targeted actions in high-risk micro-places (e.g., corner stores and vacant lots) with community-level policies aimed at disrupting systemic criminal drivers such as drug markets and socioeconomic disadvantages. In any case, the active participation of place managers, community-based organizations and residents in the design and implementation of these interventions is crucial. A positive example of the actions potentially informed by this type of analysis are the outcomes stemming from the NPSC meetings where community stakeholders proposed to lead various localized strategies to address the various issues connected to the connection between gun violence incidents and corner stores (Goldberg, 2023). For example, the Office of Violence Prevention launched a community needs assessment survey, PSE&G (New Jersey's utility company) began promoting a low-cost streetlighting system facing corner stores, while the City of Newark commits to support an increased availability of fresh food products in corner stores (NPSC, 2025).

This study is, of course, subject to several limitations. First, the absence of information and data on the actual presence of gang-related activities or drug markets reduced the possibility of verifying the supposed connection between these dynamics and gun violence events. However, qualitative insights from NPSC meetings, along with the differentiated spatial patterns observed between gun violence and aggravated assaults, lend credibility to the proposed association. Second, data on corner stores and other similar venues lack details on design features, business practices, and security measures, which could significantly impact their association with criminal behaviors. Moreover, information on the role of place managers discouraging crimes and reducing opportunities for criminal events through their presence and daily activities at specific places were not available for this study. Future research should aim to incorporate these place-specific attributes as a further level of analysis. Finally, the findings are based on data from a single urban area. The reduced extent of this geographic scope may constrain generalizability and affect the statistical power of some estimates and results. Still, despite its exploratory nature, the main contribution of this study lies in highlighting the need to examine more deeply the different criminological roles of seemingly similar micro-places and to recognize their differential impact on crime concentrations.

CRedit authorship contribution statement

Marco Dugato: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Alejandro Giménez-Santana:** Writing – original draft, Data curation, Conceptualization. **Adriana Santos:** Writing – original draft, Data curation, Conceptualization. **Joel M. Caplan:** Supervision, Conceptualization. **Leslie W. Kennedy:** Supervision, Conceptualization.

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